

Assignment 7 of Machine Learning

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1. Explain the concept of score matching and describe how it is used in score-based (diffusion) generative models.

Score matching is a technique used in generative modeling to learn the probability density function of a data distribution, $p_{\text{data}}(x)$, by instead learning its **score function**.

The Score Function

The **score function** $S(x)$ is defined as the gradient of the log-probability density function:

$$S(x) = \nabla_x \log p(x).$$

A generative model aims to parameterize a model distribution $p(x; \theta)$ and learn its parameters θ such that $p(x; \theta)$ approximates $p_{\text{data}}(x)$. The model's score function is $S(x; \theta) = \nabla_x \log p(x; \theta)$.

Why Learn the Score Function?

Directly learning the density $p_{\text{data}}(x)$ is difficult, especially because it involves an often intractable normalization constant $Z(\theta)$ in the model formulation $p(x; \theta) = \frac{1}{Z(\theta)} \exp(q(x; \theta))$.

The score function $S(x; \theta)$ is easier to work with because the intractable constant $Z(\theta)$ is independent of x and thus vanishes when taking the gradient with respect to x :

$$\log p(x; \theta) = q(x; \theta) - \log Z(\theta) \implies \nabla_x \log p(x; \theta) = \nabla_x q(x; \theta)$$

Therefore, the score function $\nabla_x \log p(x; \theta)$ does not depend on $Z(\theta)$.

Score Matching Losses

The core idea of score matching is to minimize the squared difference between the model's score function $S(x; \theta)$ and the true data distribution's score function $\nabla_x \log p_{\text{data}}(x)$.

1. Explicit Score Matching (ESM)

The **Explicit Score Matching (ESM)** objective is:

$$L_{\text{ESM}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\|S(x; \theta) - \nabla_x \log p_{\text{data}}(x)\|^2].$$

Problem: This loss requires knowledge of the true score $\nabla_x \log p_{\text{data}}(x)$, which is generally unknown.

2. Implicit Score Matching (ISM)

To bypass the need for $\nabla_x \log p_{\text{data}}(x)$, the **Implicit Score Matching (ISM)** loss is derived. Through the divergence theorem, the ESM loss can be shown to be equivalent to:

$$\mathbb{E}_{x \sim p_{\text{data}}(x)} [\|S(x; \theta)\|^2 + 2\nabla_x \cdot S(x; \theta)] + \mathbb{E}_{x \sim p_{\text{data}}(x)} [\|\nabla_x \log p_{\text{data}}(x)\|^2],$$

where $\mathbb{E}_{x \sim p_{\text{data}}(x)} [\|\nabla_x \log p_{\text{data}}(x)\|^2]$ is a constant independent of θ . Hence,

$$L_{\text{ISM}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\|S(x; \theta)\|^2 + 2\nabla_x \cdot S(x; \theta)].$$

Advantage: Minimizing $L_{\text{ISM}}(\theta)$ is equivalent to minimizing $L_{\text{ESM}}(\theta)$, but it does not require knowledge of $\nabla_x \log p_{\text{data}}(x)$. Only the model score function $S(x; \theta)$ and its divergence $\nabla_x \cdot S(x; \theta)$ are required.

Limitation: Calculating the divergence $\nabla_x \cdot S(x; \theta)$ involves second-order derivatives (the Hessian of $\log p(x; \theta)$), which can be computationally expensive and complex to implement for high-dimensional data.

Denoising Score Matching (DSM)

Denoising Score Matching (DSM) provides a practical and tractable alternative, particularly for high-dimensional data like images. It works by training the model on a perturbed (noisy) version of the data distribution, $p_{\sigma}(x)$, which is obtained by adding noise (typically Gaussian noise with variance σ^2) to the original data x_0 :

$$x = x_0 + \epsilon_{\sigma}, \quad \epsilon_{\sigma} \sim \mathcal{N}(0, \sigma^2 I).$$

The DSM objective is based on a relationship derived from the above noise process: minimizing the ESM loss for the noisy distribution $p_{\sigma}(x)$ is equivalent to:

$$L_{\text{DSM}}(\theta) = \mathbb{E}_{x_0 \sim p_{\text{data}}(x_0)} \mathbb{E}_{x|x_0 \sim p(x|x_0)} [\|S_{\sigma}(x; \theta) - \nabla_x \log p(x|x_0)\|^2],$$

where $S_{\sigma}(x; \theta)$ is the model's score function for the noisy data, and $\nabla_x \log p(x|x_0)$ is the score of the conditional noise distribution.

Key Insight (Gaussian Noise): For the common case of Gaussian noise $p(x|x_0) = \mathcal{N}(x|x_0, \sigma^2 I)$, the conditional score function has a simple closed-form solution:

$$\nabla_x \log p(x|x_0) = -\frac{1}{\sigma^2}(x - x_0).$$

Tractability: The DSM loss now only involves the simple closed form (a function of x and x_0) and the model's score function $S_{\sigma}(x; \theta)$, avoiding the complex divergence calculation needed in ISM.

How it is used in score-based (diffusion) generative models?

The Multi-Noise Score Estimation

Diffusion models work by defining a forward process that gradually adds noise (typically Gaussian) to the data x_0 over T steps (or a continuous time $t \in [0, T]$). This process creates a sequence of noisy data distributions $p_t(x)$.

The model aims to learn the true score function for each noisy distribution:

$$\nabla_x \log p_t(x) = \nabla_x \log \int p(x|x_0)p_{\text{data}}(x_0)dx_0.$$

The network $s_\theta(x, t)$ is trained to approximate this true score function for all time steps t :

$$s_\theta(x, t) \approx \nabla_x \log p_t(x).$$

Denoising Score Matching (DSM) Objective

The training is done using a variant of **Denoising Score Matching (DSM)**, which is a tractable form of score matching.

The Objective

The model is trained to minimize the expected squared difference between its predicted score $s_\theta(x, t)$ and the known true score of the conditional (noise) distribution $\nabla_x \log p(x|x_0)$:

$$L(\theta) = \mathbb{E}_t \mathbb{E}_{x_0 \sim p_{\text{data}}(x_0)} \mathbb{E}_{x \sim p(x|x_0)} [\lambda(t) \|s_\theta(x, t) - \nabla_x \log p(x|x_0)\|^2].$$

- x_0 : A data sample from the real distribution.
- t : A randomly sampled time step (noise level).
- x : The noisy version of x_0 obtained by sampling from the forward process $p(x|x_0)$.
- $\lambda(t)$: A weighting function (often $1/\sigma^2(t)$) to balance the contribution of different noise levels.

The Key to Tractability

For diffusion models, the noise added in the forward process is always Gaussian ($x = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon$, or similar), which has a simple closed-form conditional score function.

If x is generated from x_0 by adding Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma_t^2 I)$ (i.e., $x = x_0 + \epsilon$), the true conditional score is:

$$\nabla_x \log p(x|x_0) = -\frac{(x - x_0)}{\sigma_t^2} = -\frac{\epsilon}{\sigma_t^2}.$$

Since x is sampled by $x = x_0 + \epsilon$, the conditional score is proportional to the noise ϵ that was added. Therefore, the DSM loss becomes equivalent to:

$$L(\theta) \propto \mathbb{E}_t \mathbb{E}_{x_0, \epsilon} \left[\left\| s_\theta(x, t) - \left(\frac{-\epsilon}{\sigma_t^2} \right) \right\|^2 \right].$$

This formulation shows that the score network $s_\theta(x, t)$ is essentially trained as a denoiser that estimates the noise ϵ added to x , scaled by $\frac{1}{\sigma_t^2}$.

2. Unanswered Questions

DSM is highly dependent on the noise being Gaussian and isotropic ($\mathcal{N}(0, \sigma^2 I)$) for the conditional score $\nabla_x \log p(x|x_0)$ to have a simple closed form. If non-Gaussian noise (like Laplacian or Uniform noise) were used, would DSM remain equivalent? If not, what new, computable loss function would be necessary to effectively train the score network?